INCH-POUND

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SUPERSEDING
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07 October 2005

DETAIL SPECIFICATION

CLOTH, OXFORD, NYLON, 3 OUNCE

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for a three (3) ounce nylon oxford cloth.

1.2 Classification. This specification covers the following types and classes as applicable.

1.2.1 Types.

Type I - Cloth for clothing, equipage, and personnel armor (no coating).
Type II - Cloth for coating

1.2.2 Classes.

Class 1 - Undyed (natural)
Class 2 - Dyed
Class 3 - Dyed and Water Repellent Treated
Class 4 - Woodland Camouflage Pattern and Water Repellent Treated
Class 5 - Universal Camouflage Pattern (UCP) and Water Repellent Treated
Class 6 - Deleted
Class 7 - Operational Camouflage Pattern (OCP) and Water Repellent Treated

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of this address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 or 4, of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-4 - Glossary of Fabric Imperfections

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1516 - Woodland Camouflage Pattern
2-1-2519 - Universal Camouflage Pattern (UCP)
2-1-2592 - Operational Camouflage Pattern (OCP)

(Copies of drawings are available from the U.S. Army Natick Soldier Research Development and Engineering Center, ATTN: RDNS-SES-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

DOCUMENTS AND PUBLICATIONS

ENVIRONMENTAL PROTECTION AGENCY

EPA Pollutant/toxics Subtopics.

(Copies are available online at https://www.epa.gov/air-research/research-health-and-environmental.)

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FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available online at http://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1 - Gray Scale for Color Change
AATCC Evaluation Procedure 2 - Gray Scale for Staining
AATCC Evaluation Procedure 8 - AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9 - Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 22 - Water Repellency – Spray Test
AATCC Test Method 61 - Colorfastness to Laundering, Accelerated
AATCC Test Method 81 - pH of the Water Extract from Wet Processing Textiles
AATCC Test Method 96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC Test Method 118 - Oil Repellency: Hydrocarbon Resistance Test
AATCC Test Method 119 - Color Change Due to Flat Abrasion (Frosting): Screen Wire Method
AATCC Test Method 169 - Weather resistance of Textiles Xenon Lamp Exposure

(Copies are available online at http://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org.)

ASTM INTERNATIONAL

ASTM D276 - Standard Test Methods for Identification of Fibers in Textiles
ASTM D1776/D1776M - Standard Practice for Conditioning and Testing Textiles
ASTM D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)

(Copies of documents are available online at http://www.astm.org .)

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HEALTHCARE INFORMATION

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Standard sample. The dyed and finished cloth shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Basic material. The material shall be made from a 100 percent bright, multifilament, nylon yarn. It shall be an oxford weave (a plain weave with two (2) warp ends weaving as one (1)) when tested as specified in 4.6.

3.5 Color.

3.5.1 Visual shade matching (all except Class 1). The color and appearance of the finished cloth shall match the standard sample when tested as specified in 4.6. For Class 1, the cloth shall be undyed (natural).

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3.5.2 Colorfastness (all except Class 1 and Type II). The finished cloth shall conform to the colorfastness requirements listed below in Table I unless otherwise specified in the contract or procurement documents when tested as specified in 4.6.

TABLE I. Colorfastness requirements (for Type I Classes 2-5 and 7 Only).

Color Evaluation	Laundering (4 cycles) (min.)	Light (40 AFU or 170 kJ/(m^2nm)@420nm) (min.) 1/	Perspiration (acid & alkaline) (min.)	Crocking Dry/Wet (min.)	Accelerated Weathering (170 kJ/ (m^2nm) @420nm) (min.)	Frosting Carbon Black (only) (min.)
All solid colors for Classes 2 & 3	3-4	3-4	3-4	3.5	3-4	---
All colors for Classes 4 & 5 except Black 357	3-4	---	3-4	4	---	---
Black 357	3	---	3	1.0	---	4
Dk. Green 355, Brown 356, and Black 357	---	3-4	---	---	---	---
Lt. Green 354, Desert Sand 500, Urban Gray 501, and Foliage Green 502	---	3	---	---	---	---
All colors for Class 7	3-4	---	3-4	3.5	---	---
Dk. Green 528 Brown 529, And Bark Brown 561	---	3-4	---	---	---	---
Tan 525, Olive 527, Dark Cream 559, and Light Sage 560	---	3	---	---	---	---

3.5.3 Dyeing of cloth for Class 1. The cloth shall be undyed (natural).

3.5.4 Dyeing of cloth for Classes 2 and 3. The cloth shall be as specified and shall match the standard sample (see 3.2 and 6.4).

3.5.5 Dyeing/printing of cloth for Class 4 (Woodland). The cloth shall be dyed to a ground shade to match the Light Green 354 and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Light Green 354, the remaining three (3) colors shall be printed as appropriate, for the Dark Green 355, Brown 356, and Black 357 areas of the pattern. When the ground shade is not dyed to approximate Light Green 354 all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

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3.5.6 Dyeing/printing of cloth for Class 5 (UCP). The cloth shall be dyed to a ground shade matching or approximating Desert Sand 500 and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Desert Sand 500, the remaining two (2) colors shall be printed as appropriate, for the Urban Gray 501 and Foliage Green 502 areas of the pattern. When the ground shade is not dyed to approximate Desert Sand 500 all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.5.7 Dyeing/printing of cloth for Class 7 (OCP). The cloth shall be dyed to a ground shade matching or approximating Dark Cream 559 and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Dark Cream 559, the remaining six (6) colors shall be printed as appropriate, for the Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529 and Bark Brown 561 areas of the pattern. When the ground shade is not dyed to approximate Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.6 Pattern execution, Class 4, 5 and 7. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage. Skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When a standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4), and the pattern on the finished cloth shall match that of the drawing (see 2.2.2. and 6.2). Drawing number 2-1-1516 for Class 4, drawing number 2-1-2519 for Class 5 and drawing number 2-1-2592 for Class 7 shall be used. The pattern repeat for each Class shall be as follows:

Class 4 Woodland Camouflage Pattern - 27.25 (+1.25, -2.50) inches in the warp direction.
Class 5 Universal Camouflage Pattern - 36.00 (+1.25, -2.50) inches in the warp direction.
Class 7 Operational Camouflage Pattern - 25.25 (+1.25, -2.50) inches in the warp direction

3.6.1 Pattern width (filling direction for OCP, Class 7). The maximum camouflage printed area for OCP, Class 7 shall be 68-inches in the filling direction.

3.7 Spectral reflectance.

3.7.1 Classes 2 and 3, Solid Colors. The reflectance values shall conform to the requirements listed below, in Table II, when tested as specified in 4.6.

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TABLE II. Spectral reflectance requirements, Classes 2 and 3.

Reflectance Values (percent) – Solid Colors								
Wavelength, Nanometers (nm)	Coyote 498		Camouflage Green 483		Foliage Green 504		Tan 499	
	Min.	Max.	Min.	Max.	Min.	Max.	Max.	Min.
600	8	20	3	10	8	26	8	26
620	8	20	3	10	8	26	8	26
640	8	22	3	10	8	28	8	30
660	8	24	3	11	10	30	8	34
680	12	24	3	13	10	34	12	38
700	12	34	4	28	12	38	12	40
720	16	42	5	40	16	42	16	46
740	22	46	7	52	16	46	22	50
760	30	50	11	60	18	48	30	50
780	34	54	17	64	18	48	34	54
800	36	56	24	67	20	50	36	56
820	38	58	32	70	22	54	38	58
840	38	58	37	71	24	54	38	58
860	40	60	40	73	26	56	40	60

3.7.2 Class 4, Woodland Camouflage. The reflectance values shall conform to the requirements listed below, in Table III, when tested as specified in 4.6.

TABLE III. Spectral reflectance requirements, Class 4.

Reflectance Values (percent) - Woodland Camouflage						
Wavelength, Nanometers (nm)	Light Green 354		Dark Green 355 and Brown 356		Black 357	
	Min.	Max	Min.	Max	Min.	Max
600	8	20	3	9	N/A	N/A
620	8	20	3	9	N/A	N/A
640	8	20	3	9	N/A	N/A
660	8	20	3	12	N/A	N/A
680	10	30	3	16	N/A	N/A
700	18	50	5	32	N/A	20
720	22	54	7	44	N/A	30
740	30	56	12	52	N/A	33
760	35	58	18	56	N/A	33
780	40	62	26	56	N/A	34
800	55	80	34	56	N/A	34
820	55	80	42	60	N/A	35
840	55	84	44	60	N/A	35
860	60	84	44	60	N/A	35

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3.7.3 Class 5, Universal Camouflage. The reflectance values shall conform to the requirements listed in Table IV, when tested as specified in 4.6.

TABLE IV. Spectral reflectance requirements, Class 5.

Reflectance Values (percent) - Universal Camouflage Pattern						
Wavelength, nanometers (nm)	Desert Sand 500		Urban Gray 501		Foliage Green 502	
	Min	Max	Min	Max	Min	Max
600	28	40	12	26	8	18
620	30	42	14	26	8	18
640	34	48	14	28	8	20
660	38	56	14	30	10	26
680	44	60	18	34	10	26
700	46	66	24	38	12	28
720	48	68	26	42	16	30
740	48	72	30	46	16	30
760	50	74	32	48	18	32
780	54	76	34	48	18	34
800	54	76	34	50	20	36
820	54	76	36	54	22	38
840	55	78	38	54	24	40
860	56	78	40	56	26	42

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3.7.4 Class 7, Operational Camouflage Pattern (OCP). The reflectance values shall conform to the requirements listed in Table V, when tested as specified in 4.6.

TABLE V. Spectral reflectance requirements, Class 7.

Reflectance values (percent) - Operational Camouflage Pattern (OCP)						
	Dark Cream 559 & Tan 525		Light Sage 560, Olive 527 & Brown 529		Dark Green 528 & Bark Brown 561	
Wavelength (nm)	Min	Max	Min	Max	Min	Max
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.8 Physical requirements. The finish cloth shall conform to the physical requirements listed below, in Table VI, when tested as specified in Table 4.6.

TABLE VI. Physical requirements.

Characteristic	All Types & Classes
Weight, oz./sq.yd. (minimum)	3.0
Yarns per inch, (minimum) Warp Filling	 180 76
Breaking strength (pounds), (minimum) Warp Filling	 220 135

3.8.1 Width. For government procurements only, the width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.9 Finish. The type of finish shall conform to that shown by the standard sample and the requirements stated below for that specific Type and Class.

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3.9.1 Type I (Classes 1, 2, and 3). The cloth shall be lightly calendered. Class 3 shall be water repellent treated and it shall conform to the requirements specified in 3.10 and 3.11.

3.9.2 Type I (Classes 4, 5, and 7). The cloth shall be printed (see 6.2) and meet the requirement of this document for camouflage patterns. The cloth shall be water repellent treated and it shall conform to the requirements specified in 3.10 and 3.11.

3.9.3 Type II. The cloth shall be heat set and meet the requirements of 3.12, but not calendered. Coating requirements shall be as specified in the contract or purchase order (see 6.2).

3.10 Spray rating (Type I, Classes 3, 4, 5 and 7). The results of the three (3) individual determinations on the water repellent treated cloth sample for spray rating shall be a minimum of 70, 70, 70 after three (3) laundering, when tested as specified in Table 4.6.

3.11 Resistance to organic liquid (Type I, Classes 3, 4, 5 and 7). The cloth shall not show wetting by n-tetradecane initially and after three (3) laundering, when tested as specified in 4.6.

3.12 Dimensional stability. The cloth shall show no appreciable distortion or puckering. There shall not be more than 2.0 percent dimensional change in either warp or filling direction when tested as specified in 4.6.

3.13 pH. The pH value of the finished cloth shall be no lower than 5.0 nor higher than 8.5 when tested as specified in 4.6.

3.14 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used (see 2.3).

3.14.1 Toxicity documents. All finishes/chemicals used to process the garment shall be identified and accompanied by the appropriate Material Safety Data Sheet (MSDS) information. The use of chemicals recognized by the Environmental Protection Agency (EPD) as human carcinogens is prohibited.

3.15 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put up in a full width on a roll as specified in 5.1.

3.16 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.17 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word “FACE” at each end of the roll.

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3.18 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the applicable established maximum point values.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1.1, shall be inspected for conformance with physical requirements of 4.6, examined for appearance, color, defects in Table VIII, and overall workmanship. The presence of excessive defects, as defined in the contract (see 6.2) or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection, submitted in accordance with 3.1.2, shall include the examination of 4.5 and the tests of 4.6 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with acceptance quality levels as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2).

4.5 Yard-by-yard examination. Each roll in the sample shall be examined yard-by-yard on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.5.1. No linear yard (increments of one (1) yard on the measuring device of the inspection machine) from any one (1) roll within the sample shall be penalized more than four (4) points. Acceptance criteria for visual examination shall be as specified in the contract or purchase order (see 6.2). The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one (1) roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case rolls from all shipping containers shall be present in the sample.

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Point computation for lot quality and individual roll quality shall be as follows:

Total points scored in sample x 3600 = Points per 100 square yards Contracted width of cloth (inches) x Total yards inspected

4.5.1 Demerit points. Demerit points shall be assigned as follows:

For defects three (3)-inches or less in any dimension	- one (1) point
For defects exceeding three (3)-inches but not exceeding six (6)-inches in any dimension	- two (2) points
For defects exceeding six (6)-inches but not exceeding nine (9)-inches in any dimension	- three (3) points
For defects exceeding nine (9)-inches in any dimension	- four (4) points

4.5.2 Visual examination. The cloth shall be examined for the defects listed in Table VII.

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TABLE VII. Visual examination defects.

Examine	Defect
Material defects and damages	Baggy, ridgy or wavy cloth, any hole, cut, tear smash, thin place, visible mends, broken or missing yarn. Uneven weaving throughout. Area of poor dye penetration, dye streak, or shade bar Colors/print not as specified. Marking on face side of cloth omitted. Edge ravels when pulled outward Slack or tight selvage 1/ Width not as specified within a tolerance of ±1/2-inch. Camouflage pattern (face side) Any skitteriness of pattern exceeding that shown by the standard sample. Pattern design not equal to standard sample. Excessive feathering or spew of pattern. Pattern repeat not equal to the standard sample. arpwise pattern repeat: Woodland Camouflage pattern (Class 4) less than 24.75-inches or more than 28.50-inches. Universal Camouflage pattern (Class 5) less than 33.50-inches or more than 37.25-inches. Operational Camouflage pattern (Class 7) less than 22.75-inches or more than 26.50-inches. Pattern width (Operational Camouflage pattern ONLY) more than 68-inches in the filling direction. Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample. Excessive haloing or trapping (overlapping of colors) of pattern as compared to the standard sample. Any color off shade, not uniform, mottled, or spotted (face side only) Overall application of water repent not uniform Tackiness (sticky to touch)
Cleanliness	Any spot, streak, or stain of a permanent nature. Any embedded foreign matter in the cloth. 2/ Objectionable odor.
Toxicity documents	Material Safety Data Sheets (MSDS) missing

1/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed, and the finished cloth examined for the selvage conditions. Suspect rolls shall be removed from the perch and unrolled on the floor or table and further examined for the conditions. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.
2/ Clearly visible at normal inspection distance 3-feet.

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4.5.3 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined that each roll is labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act (see 3.16).

4.6 End item testing. The cloth shall be tested for the characteristics listed in Table VII. The methods of testing wherever applicable and as listed in Table VII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more sample units fail to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.6.1 Basic cloth. The basic cloth shall be tested in accordance with Table VII.

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TABLE VIII. End item tests.

Characteristic	Requirement paragraph	Test method
Nylon Identification	3.4.1	ASTM D276 or AATCC 20 (see 6.5)
Luster	3.4.1	Visual 1/
Multifilament	3.4.1	Visual 1/
Weave	3.4.1	Visual 1/
Visual shade matching: (Classes 2 - 5 and 7)	3.5.1	4.7.2
Colorfastness: (Classes 2-5 and 7)	Table I	
Light		AATCC 16.3, Option 3 2/
Laundering after four (4) cycles		AATCC 61, Test 3A 3/
Crocking		AATCC 8 4/
Perspiration (acid and alkaline)		AATCC 15 3/
Accelerated weathering		AATCC 169, Option 4 2/
		AATCC 119 2/
Spectral reflectance:		4.7.5
Classes 2 and 3	3.7.1	
Class 4	3.7.2	
Class 5	3.7.3	
Class 7	3.7.4	
Weight	Table VI	ASTM D3776/D3776M
Yarns per inch	Table VI	ASTM D3775
Breaking Strength	Table VI	ASTM D5034
Calendared: (Type I, Classes 1,2, and 3)	3.9.1	Visual 1/
Water repellent treatment: (Type I, Classes 3, 4, 5 and 7)		
Spray rating		
After three (3) launderings	3.10	AATCC 96, Test VIc and AATCC 22
Resistance to organic liquids:		
Initially	3.11	AATCC 118
After three (3) launderings	3.11	AATCC 96, Test VIc and AATCC 118
Dimensional stability		
After one (1) cycle	3.12	AATCC 96 VIc 5/
pH	3.13	AATCC 81
Toxicity	3.14	4.8

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ Rated using AATCC Evaluation Procedure 1, Gray Scale for Color Change

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3/ Rated using AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining
4/ Rated using AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
5/ The dimensional stability shall be performed after one (1) cycle. The cloth shall not be pressed after tumble drying prior to measurement.

4.7 Methods of testing and inspection. All testing shall be done in a standard condition environment defined by ASTM D1776/D1776M, if not specifically defined by the individual test procedure.

4.7.1 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicate which colors were evaluated together.

4.7.2 Visual shade matching (Classes 2-5 and 7). The color and appearance of the cloth shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (±200) illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (±200).

4.7.3 Operational Camouflage Pattern (OCP) (Class 7). All colorfastness shall be performed on the solid color area and not the tonal area (see Table I).

4.7.4 Pattern execution. The pattern of the cloth shall be matched to the specified pattern drawing (see 2.2.2).

4.7.5 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for Classes 2-5 and 7 at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The cloth for Class, 2, 3, and 4, shall be measured as a single layer using three (3) backing layers of the same shade for Light Green 354, Dark Green 355 and Brown 356 colors. The cloth for Classes 5 and 7 shall be measured as a single layer backed with four (4) layers of the same shade. The specimen shall be viewed at an angle no greater than 10 degrees from normal, with the specular component included. Measurements shall be taken on a minimum of two (2) different areas. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns when presented to the sample port. Photometric accuracy of the spectrophotometer shall be within 1-percent and wavelength accuracy within 2-nanometers. The diameter for standard aperture size used in the color

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measurement device shall be 1.0 to 1.25-inches for Class 2, 3, and 4, and 0.3725-inches or larger for the Classes 5 and 7. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.8 Toxicity test. Unless otherwise specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.14) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Type I cloth is intended for use in clothing, equipage and personnel armor. The Type II cloth is for coating purpose to be used as applicable.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and Class of cloth required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2)
d. Color/print of cloth required (see 3.5.3 – 3.5.7)
e. When first article is required (see 3.1.1, 4.2 and 6.3)
f. Camouflage pattern drawing if required (see 3.6).
g. Width of cloth required (see 3.8.1).
h. Conformance inspection acceptance quality limits (AQL) (see 4.3)
i. When toxicity testing is required (see 3.14 and 4.8)
j. Packaging (see 5.1)

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6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Coating Equipage Operational
Operational Camouflage Pattern – (OCP)
Personnel armor
Printed Universal
Water repellent treated Woodland

Custodians:
Army - GL
Navy - NU

Review activities:
Army – MD
Navy – AS, MC
Air Force - 11

Preparing activity:
DLA - CT

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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